R18

Code No: 153AB

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, August/September - 2022
ANALOG AND DIGITAL ELECTRONICS
(Common to CSE, IT, ECM, ITE, CSE(SE), CSE(CS), CSE(N))

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Discuss the different types of junction breakdowns that can occur in a reverse biased diode.
b) Explain operation of diode in forward bias and reverse bias condition. Draw V-I characteristics of diode. [8+7]

2.a) Explain about the full-wave center-tap rectifier with L section filters and also draw suitable diagram and waveforms.
b) Write a short note on diffusion capacitance and diode switching times. [8+7]

3.a) With neat diagrams and necessary equations, explain the effect of coupling capacitor and bypass capacitor on the performance of an amplifier at low-frequencies?
b) Explain how self-biasing can be done in a BJT with relevant sketches and waveforms? [8+7]

4.a) Write equations of voltage gain, current gain, Input impedance and Output impedance of CE amplifier.
b) Explain how transistor acts as an amplifier? [10+5]

5.a) Bring out the differences between BJT and FET. Compare the three configurations of JFET amplifiers.
b) With the help of a neat schematic, explain the functioning of a common drain amplifier. [8+7]

6.a) Define the De Morgan Laws with suitable example.
b) Implement the basic logic gates by using modified DTL gates, HTL and TTL gates. [6+9]

7.a) Explain the function of a Encoder with necessary diagrams and discuss its applications.
b) Design the 4-bit binary Adder-Subtractor with suitable diagram. [6+9]

8.a) Give the design of 3 bit Ring counter and explain its operation with waveforms. Also give the applications of ring counter.
b) Obtain the characteristic equations of D and T flip flops. [9+6]

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